

# Analysis of Quantum Vulnerable Bitcoin



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# MOTIVATION

How much Bitcoin is immediately quantum vulnerable (QV) and why?

*“5M-10M BTC”*

*“6,262,905 BTC”*

?

*“About 25%”*

*“now closer to 20%”*

# OUTLINE

**Breakdown of sources of QV, including over time**

1. **Address reuse, revealed keys/scripts**
2. **Inherently vulnerable script types**
3. **Fork exposure, with a focus on Bitcoin Cash (BCH)**

**Vulnerability assessment of the top 1000 addresses**

**Any implications for migration efforts, burn vs steal decision?**

# ~6.51M BTC IS IMMEDIATELY QV

32.7% of total supply

Address Reuse: 4.49M BTC (69%)

Inherently Vulnerable: 1.87M BTC (28.8%)

- P2PK: 1.72M
- P2TR: 0.15M
- P2MS: 69 only!

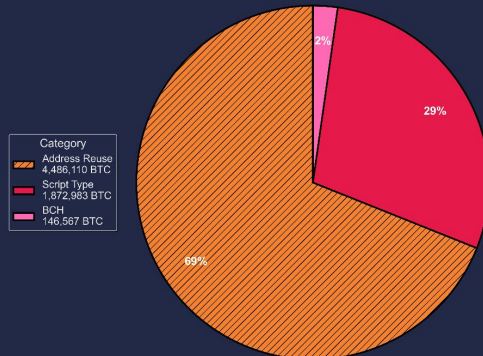
BCH Fork Exposure: 0.15M BTC (2.3%)

*\*As of block height 900,000 (June 6, 2025)*

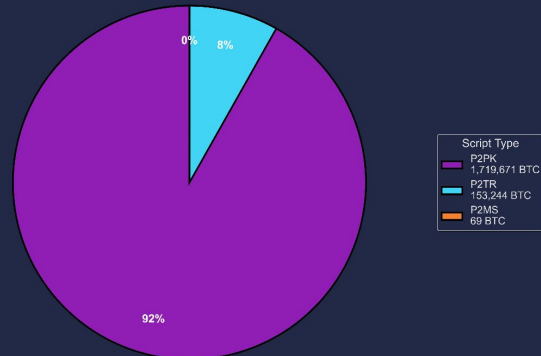
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Quantum Vulnerability Breakdown at Block 900,000

High-Level Vulnerability Categories



Inherently Vulnerable Script Types



# HISTORICAL TRENDS AND BCH FORK EXPOSURE

Reconstruct UTXO set every 50K blocks from block height 400K to 900K

- UTXOs associated with reused addresses
- UTXOs belong to inherently vulnerable scripts
- Generally understand UTXO set composition at snapshots

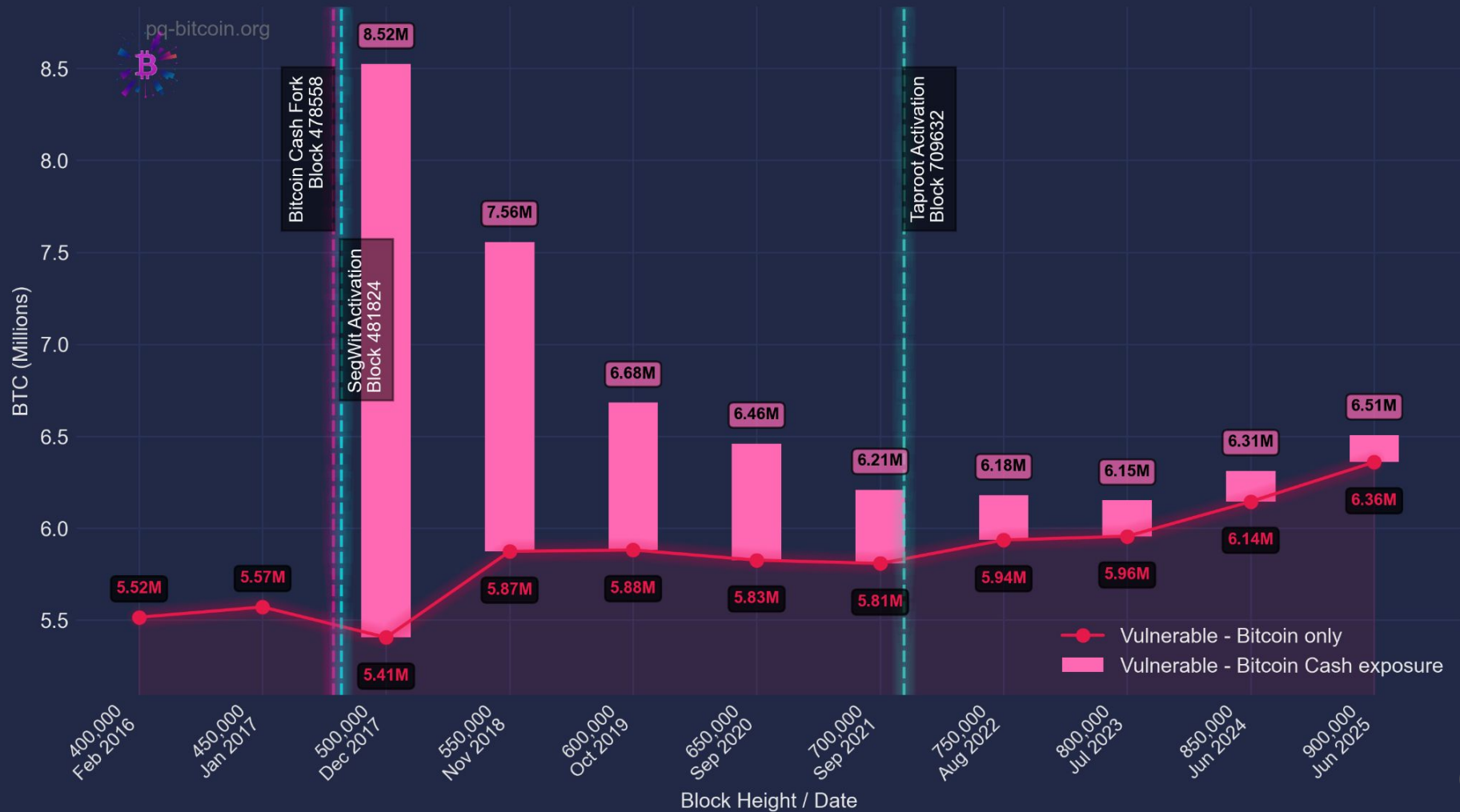
## BCH Fork Exposure

- Post fork (478558), determine which TX outputs have been spent on BCH but remain unspent on Bitcoin
- Only consider the marginal contribution of BCH spends

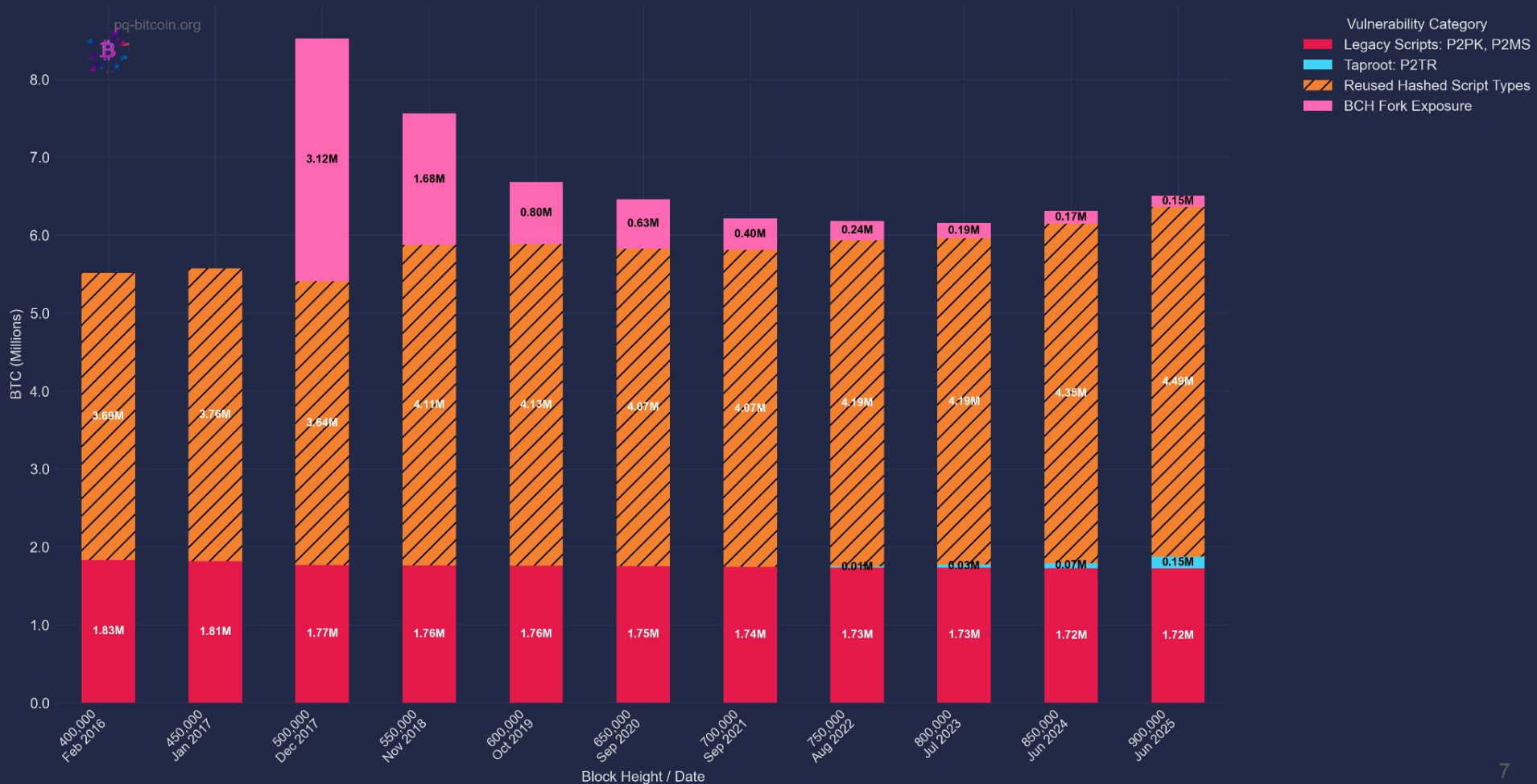


*Analysis performed on Google's BigQuery, using [BTC](#) and [BCH](#) datasets*

# Quantum Vulnerable BTC Over Time

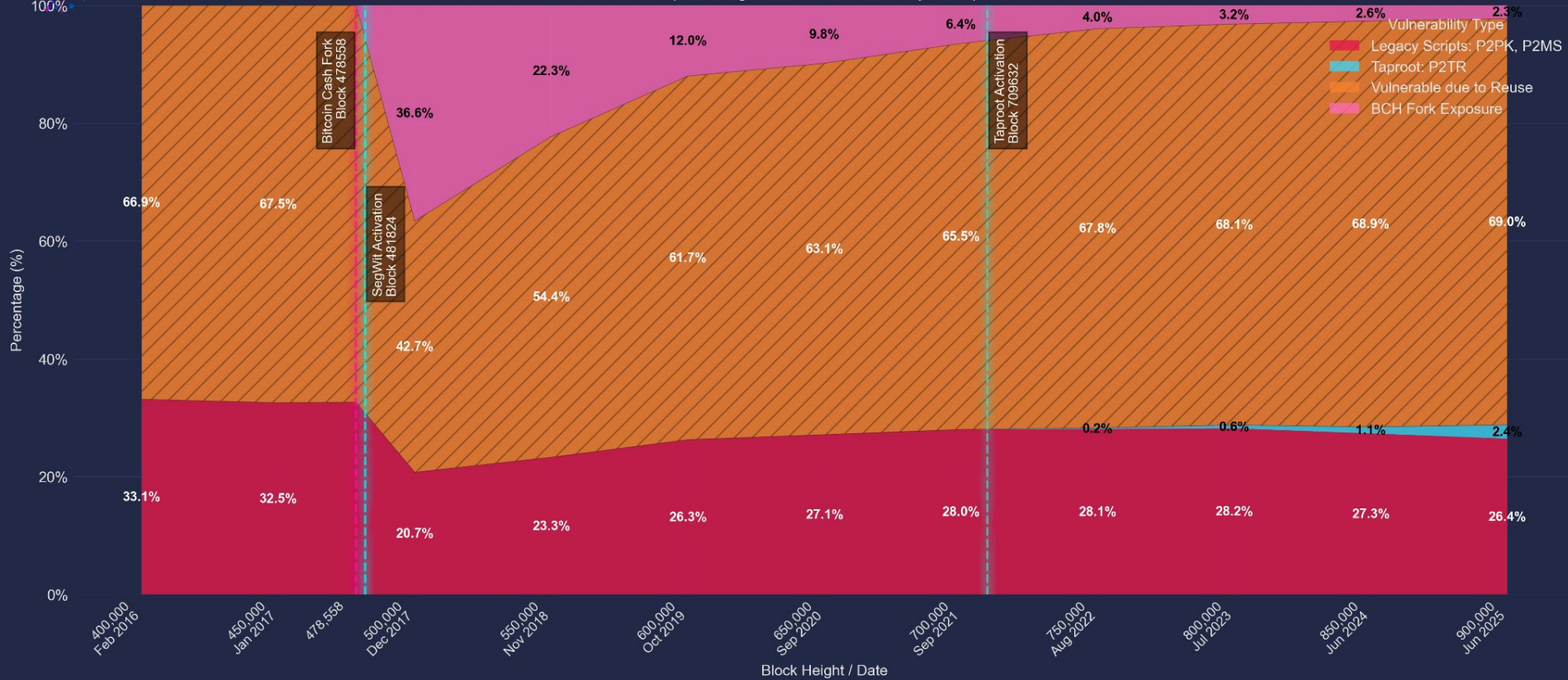


# Breakdown of Quantum Vulnerable BTC by Category





# Quantum Vulnerability Sources by Percentage (Including Bitcoin Cash Fork Exposure)





# ADDRESS REUSE CAVEATS

Not all revealed P2SH/P2WSH scripts are immediately vulnerable, could be protected by non-signature mechanisms:

- Unexpired timelocks - Protection remains until time-lock expires
- Unused hash locks - Only if preimage hasn't been revealed in any prior spend
- Complex conditionals - Where non-hash paths were used

We assume that a majority of P2SH/P2WSH are immediately vulnerable (e.g. they are multisig), and leave analysing the intricacies for future work.

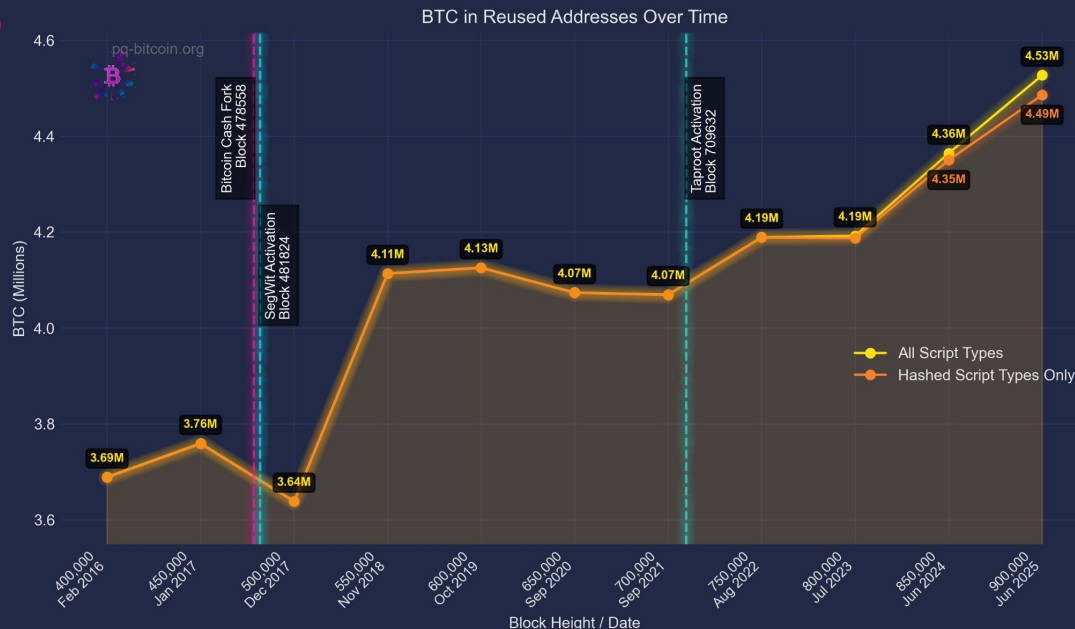
# ADDRESS REUSE

500→550K: Dec 2017→Nov 2018

- Bitcoin's first mainstream adoption wave
- Retail and exchanges?

800→900K: Jul 2023→June 2025

- ETFs (Jan 2024) & increased financialization?

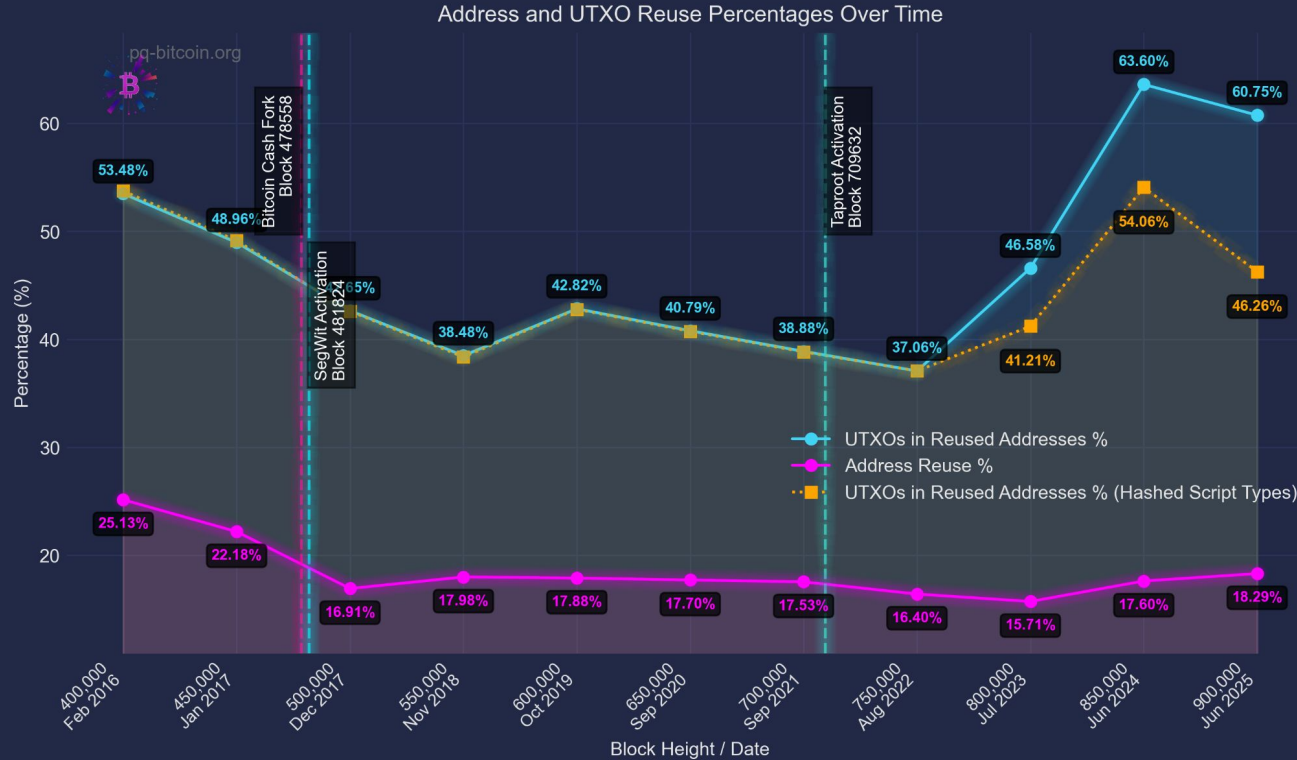


# ADDRESS REUSE

Unique address:  
addresses that have at  
least one UTXO at  
each specific block height

**Address Reuse %** is the  
% of unique addresses  
that have been reused

**UTXOs in Reused  
Addresses %** is the %  
of UTXOs that belong  
to a reused address



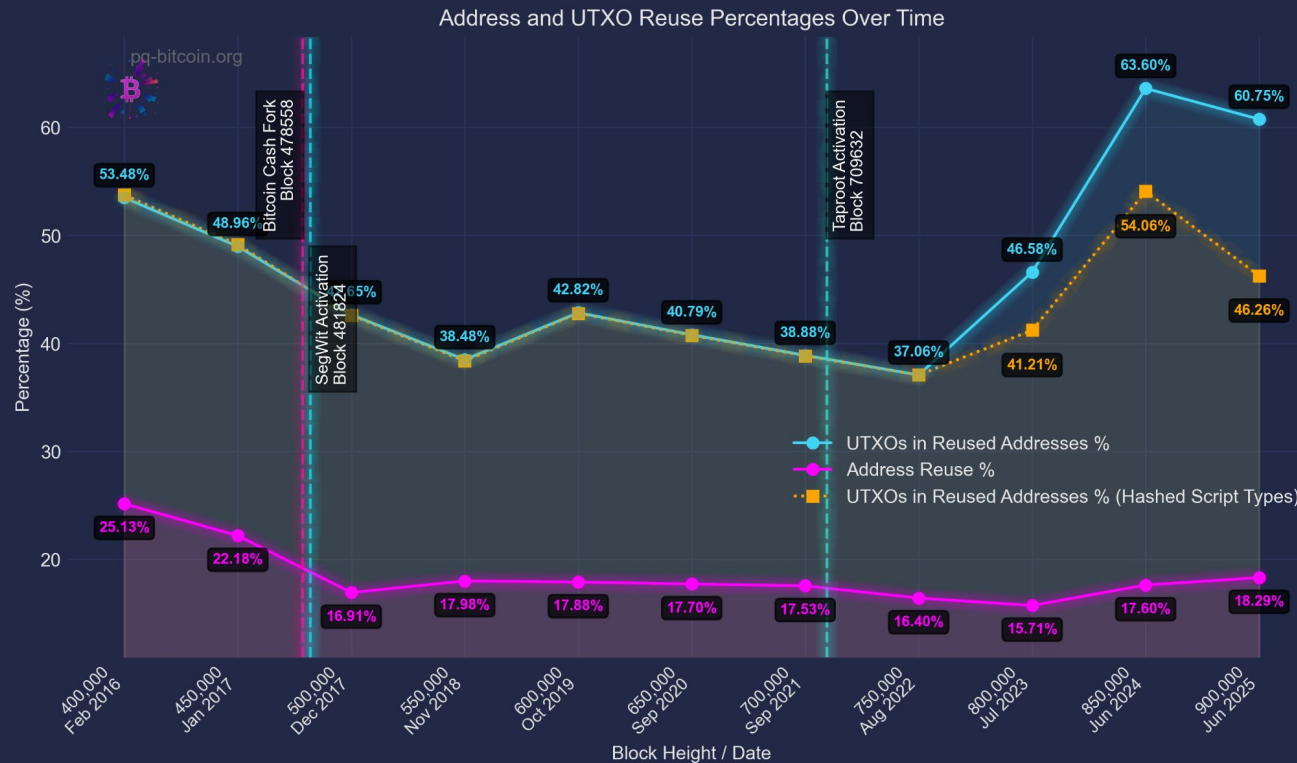
# ADDRESS REUSE

## Currently:

- < 20% addresses are reused...
- ... but  $\frac{2}{3}$  UTXOs belong to reused address

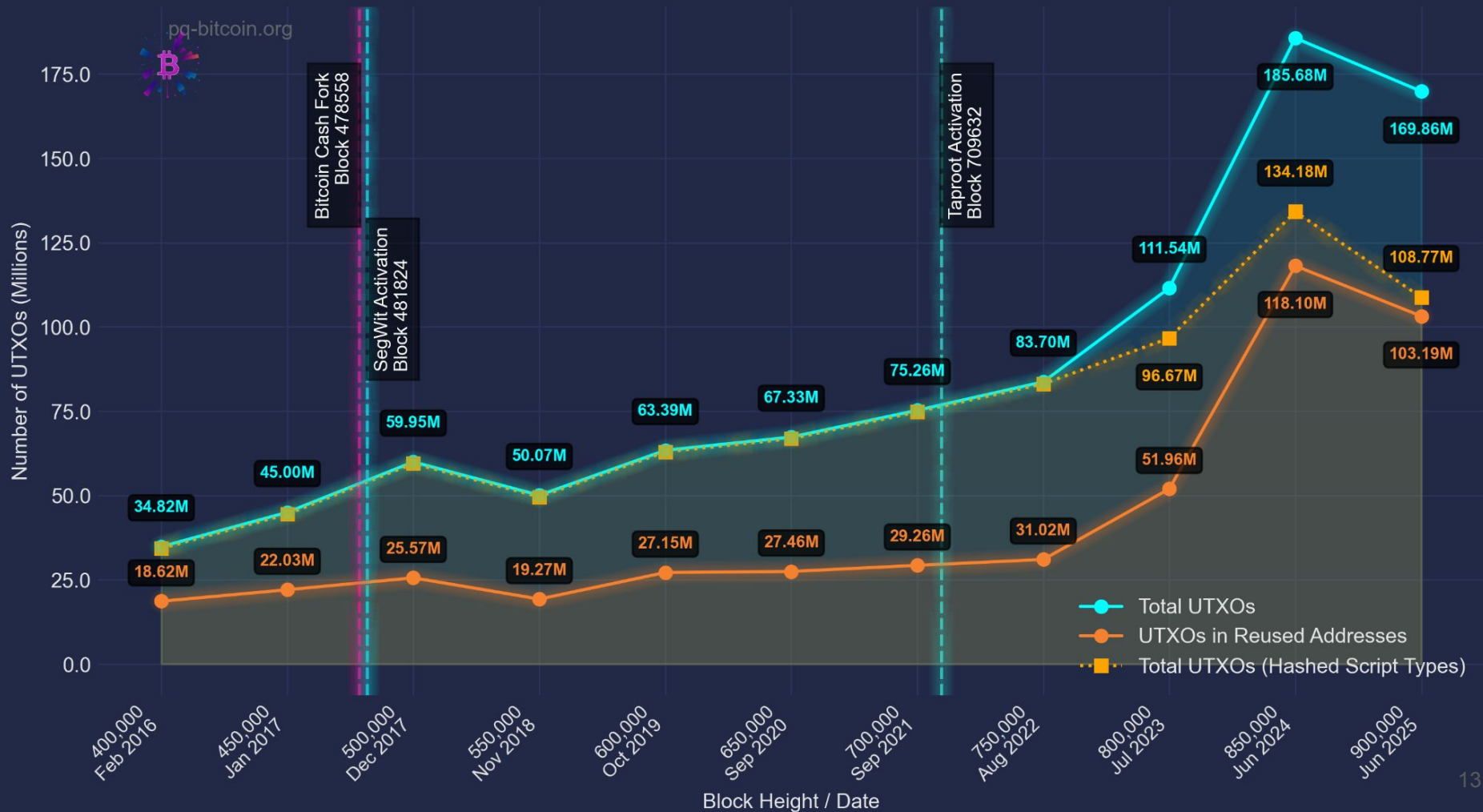
*A majority of addresses (80-85%) are single-use*

*A minority of addresses (15-20%) are reused and account for a majority of UTXOs.*



# Total UTXOs vs. UTXOs in Reused Addresses Over Time

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# TOP 1000 ADDRESSES ANALYSIS

## Top 1000 addresses by balance:

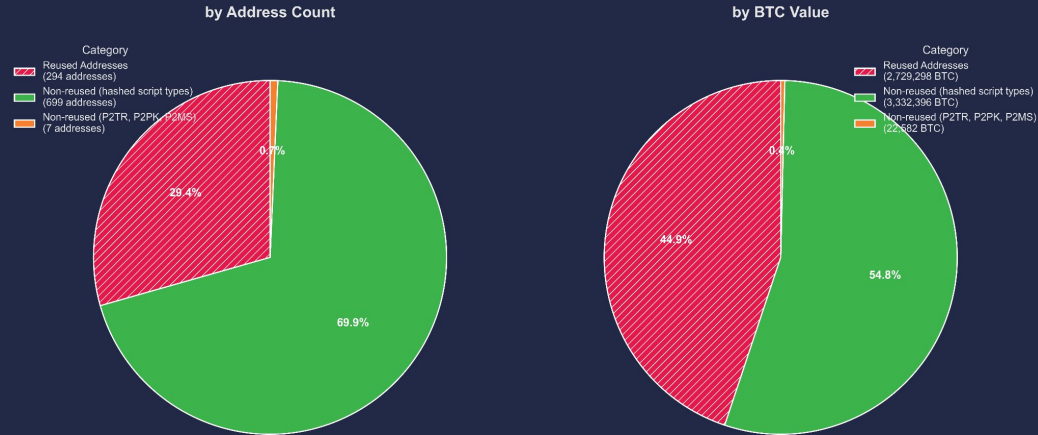
- ~6.08M BTC
- ~31% of total supply
- 1<sup>st</sup> address: 248,598 BTC (Binance)
- 1000<sup>th</sup> address: 1,738 BTC (no label)

## Reuse:

- 294 addresses
- ~2.73M BTC (~45% of T1000)

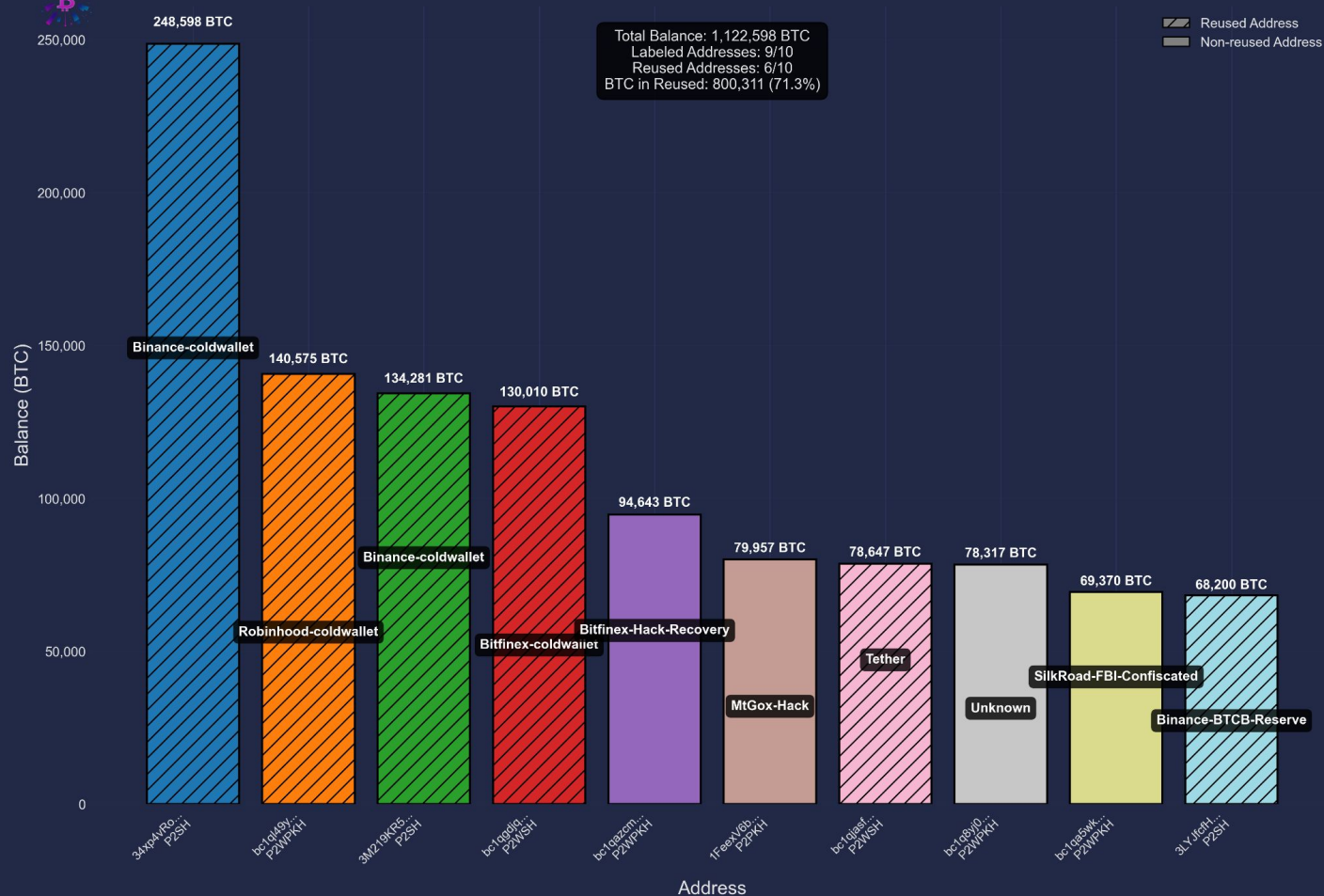
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Breakdown of Address Reuse Status in Top 1000 Addresses

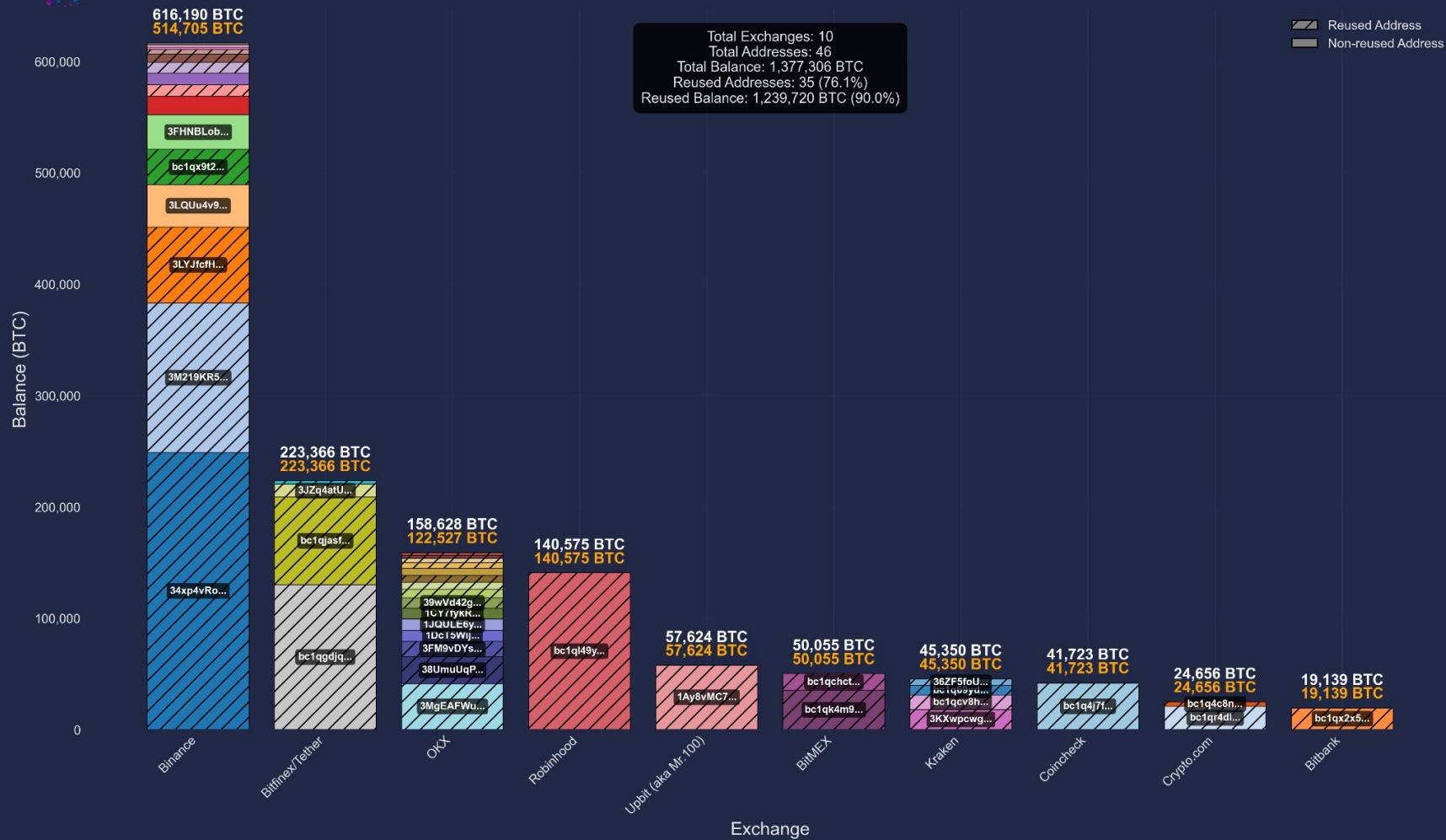


Data sourced from [BitInfoCharts' Rich List](#) (including address labels)

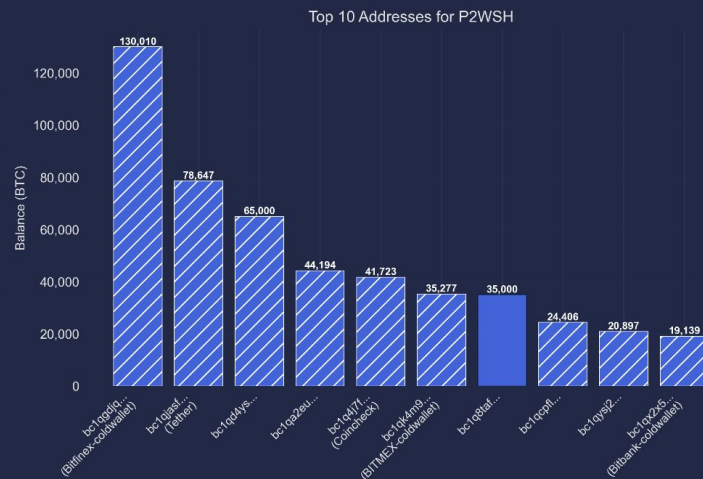
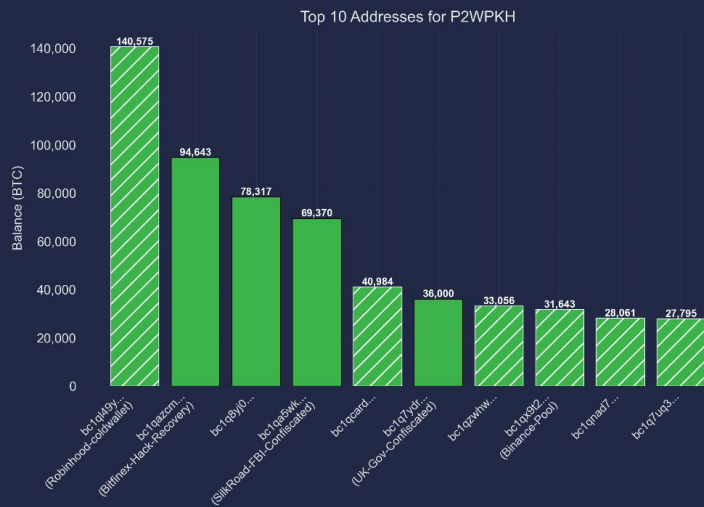
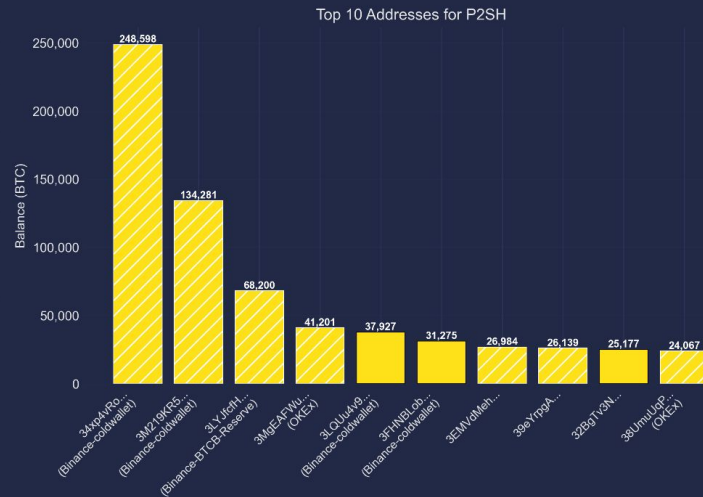
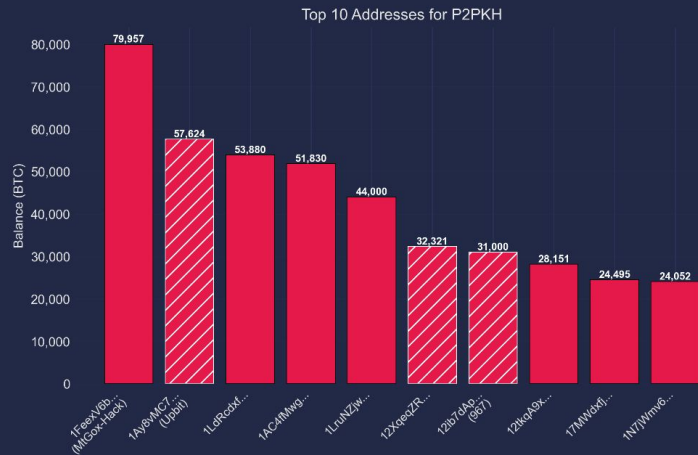
\* As of block height ~900,500 (June 10, 2025)





Top 10 Exchange Entities by Balance  
(Stacked by Address)

## Top 10 Addresses by Hashed Script Type



# IMPLICATIONS FOR MIGRATION, BURN/STEAL

We have 3 main sources of immediately QV Bitcoin,  
what do they mean for mean for migration, burn/steal

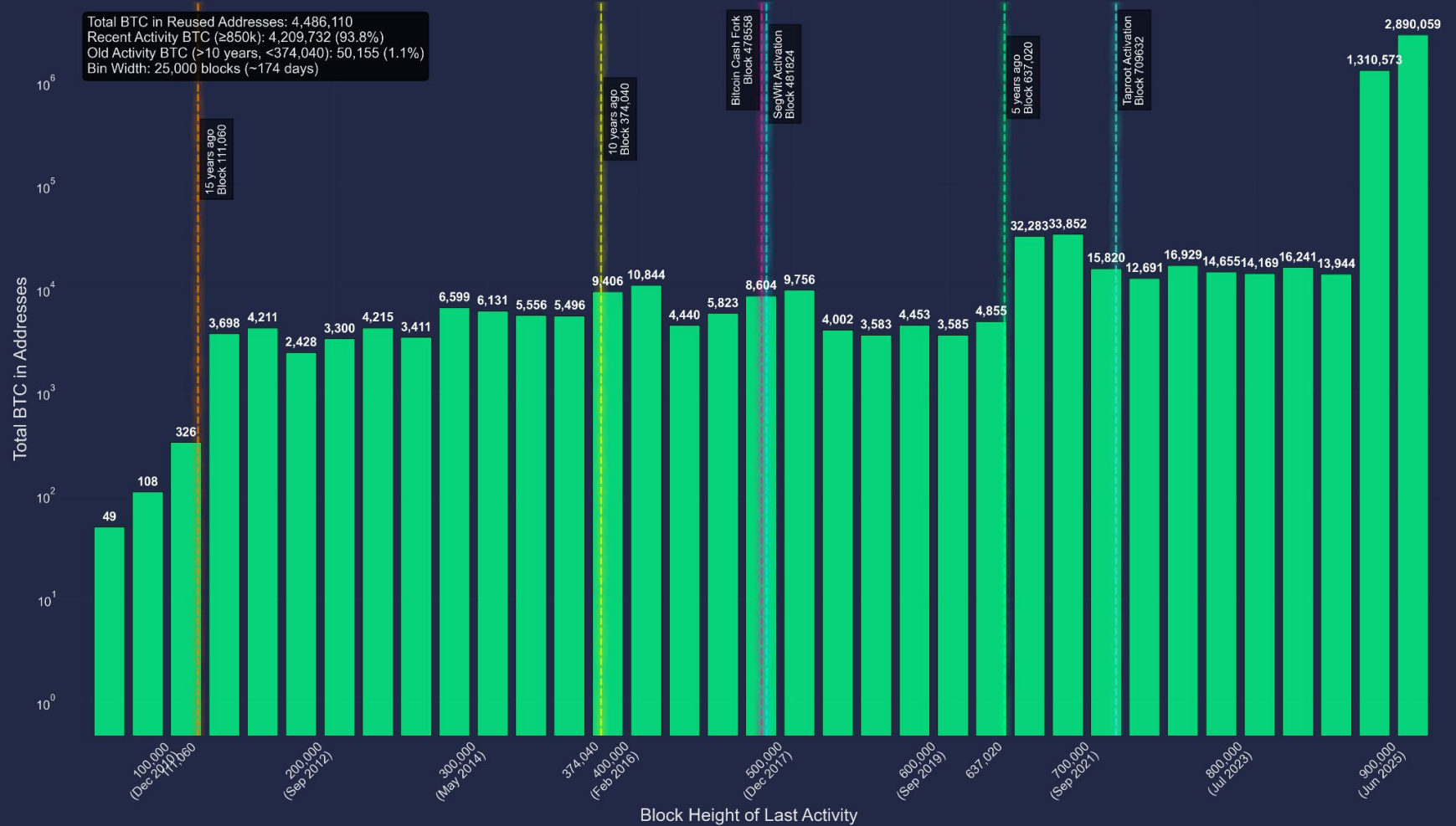
BCH fork exposure:

- Spent on BCH within last 8 years → can still be moved today?
- ∴ Most BCH fork exposure UTXOs can be moved? ~150K BTC

Reused addresses:

- How many reused address UTXOs can be moved?
- Consider last activity of associated addresses

# Last Activity Distribution of Reused Bitcoin Addresses (BTC Value)



# REUSED ADDRESSES

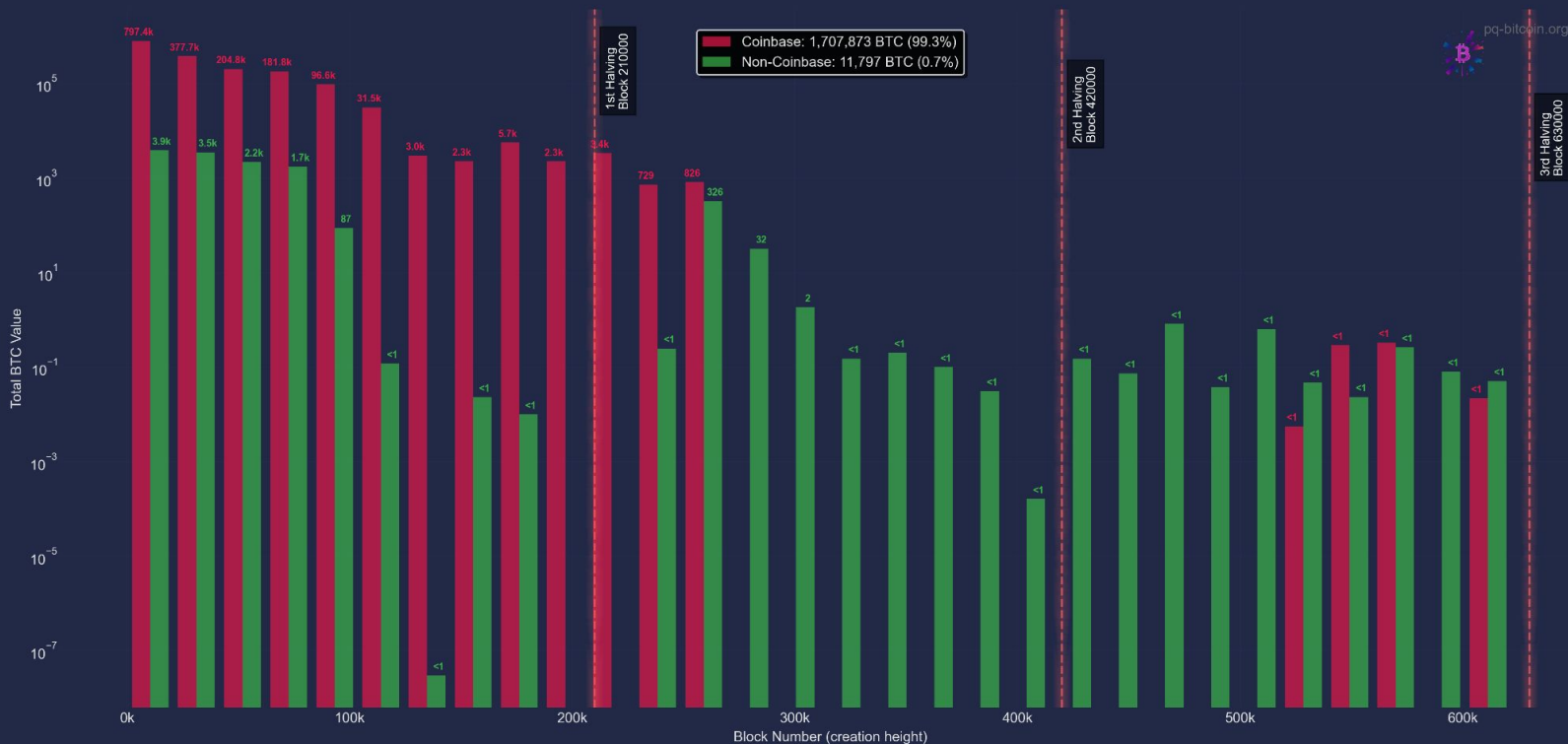
| Activity Period         | Addresses | Address % | Total BTC  | BTC % |
|-------------------------|-----------|-----------|------------|-------|
| Within last 5 years     | 5354856   | 63.73     | 4371782.98 | 97.45 |
| Between 5 and 10 years  | 2224323   | 26.47     | 64172.36   | 1.43  |
| Between 10 and 15 years | 823216    | 9.80      | 49784.88   | 1.11  |
| More than 15 years      | 221       | 0.00      | 370.20     | 0.01  |

**Funds associated with reused addresses that we might consider lost (from a last activity perspective) are negligible → can still be moved today?**

# INHERENTLY VULNERABLE SCRIPTS - P2PK

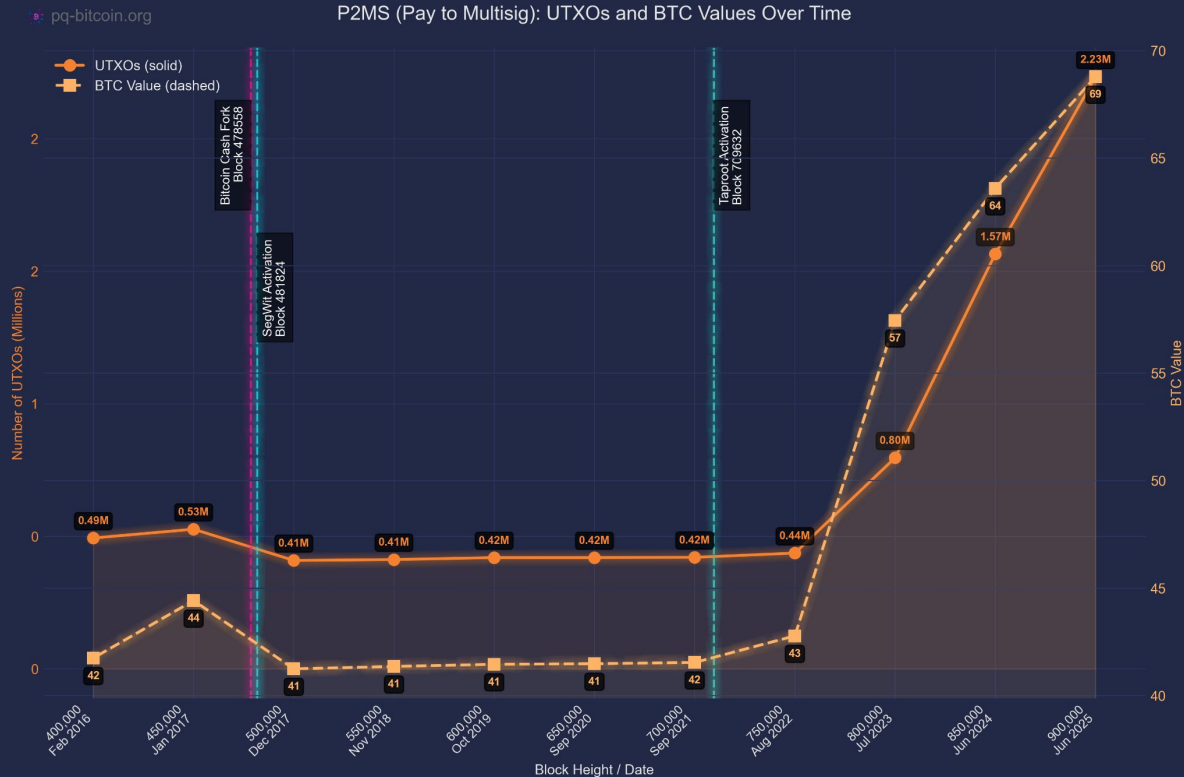
New UTXOs not being created, but significant historical BTC value

BTC Value Distribution by Age: Coinbase vs Non-Coinbase



# INHERENTLY VULNERABLE SCRIPTS - P2MS

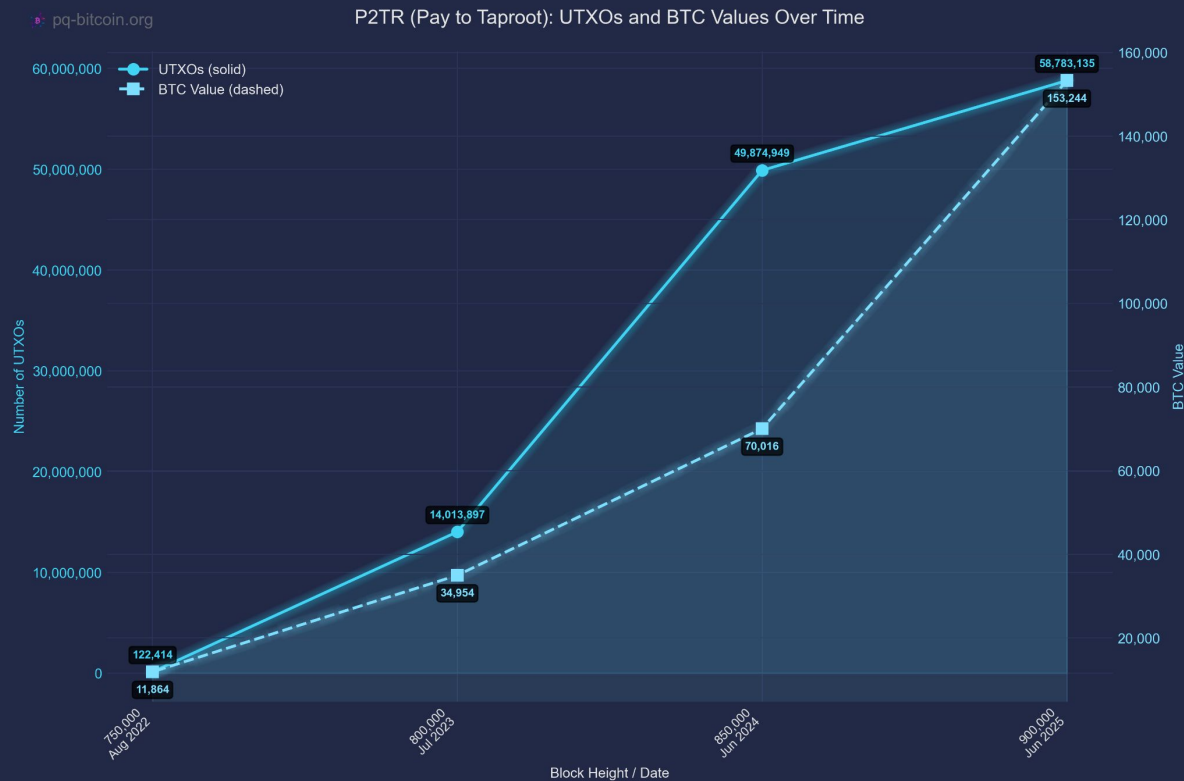
New UTXOs are being created today, but miniscule BTC value



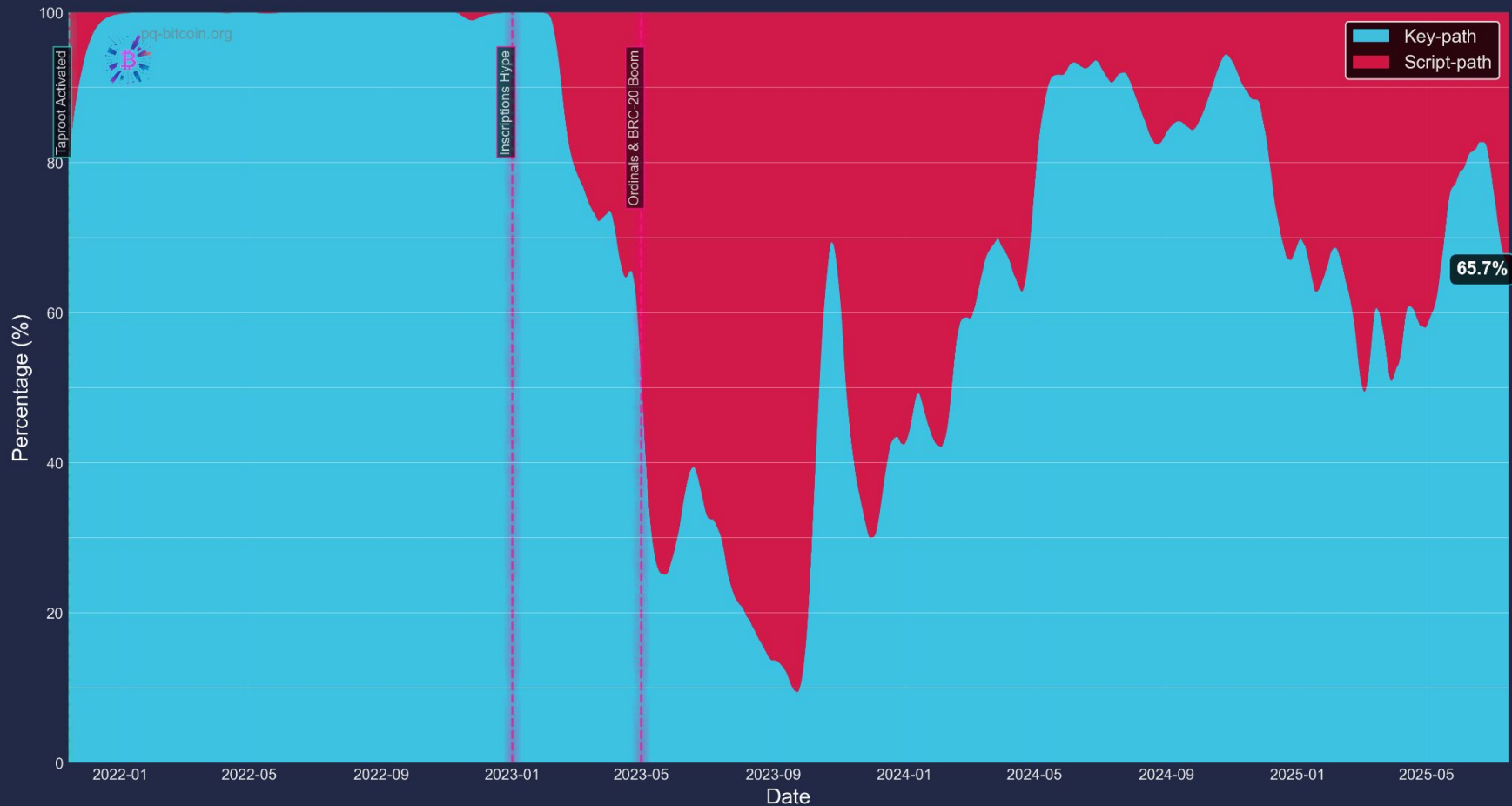


# INHERENTLY VULNERABLE SCRIPTS - P2TR

Growing in both # of UTXOs and BTC encumbered



## P2TR (Pay to Taproot): Key-path vs Script-path



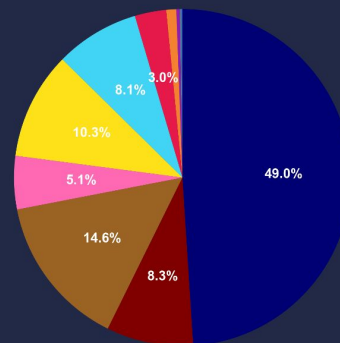
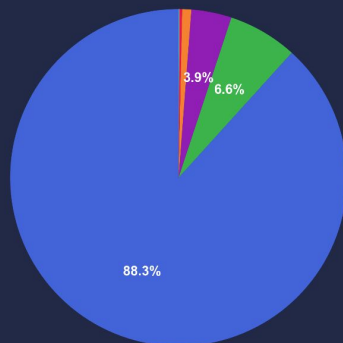


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Count

## P2TR UTXO Distribution by Size Range

Value



### P2TR Distribution Summary

| Size Range       | UTXO Count | Count % | Value (BTC) | Value % |
|------------------|------------|---------|-------------|---------|
| 0 - 1k sats      | 51,218,283 | 88.3%   | 248.9       | 0.2%    |
| 1k - 10k sats    | 3,855,459  | 6.6%    | 114.3       | 0.1%    |
| 10k - 100k sats  | 2,242,806  | 3.9%    | 545.8       | 0.3%    |
| 0.001 - 0.01 BTC | 505,716    | 0.9%    | 1.5k        | 1.0%    |
| 0.01 - 0.1 BTC   | 145,261    | 0.3%    | 4.7k        | 3.0%    |
| 0.1 - 1.0 BTC    | 40,064     | 0.1%    | 12.6k       | 8.1%    |
| 1 - 5 BTC        | 8,029      | 0.0%    | 16.1k       | 10.3%   |
| 5 - 10 BTC       | 1,169      | 0.0%    | 8.0k        | 5.1%    |
| 10 - 50 BTC      | 1,223      | 0.0%    | 22.9k       | 14.6%   |
| 50 - 100 BTC     | 185        | 0.0%    | 13.0k       | 8.3%    |
| > 100 BTC        | 187        | 0.0%    | 76.6k       | 49.0%   |

Total P2TR UTXOs: 58,018,382  
Total P2TR BTC: 156,336.63



# IMPLICATIONS FOR MIGRATION, BURN/STEAL

P2PK - The main problem (we all knew this)

- ~1.72M BTC, 8.65% of total current BTC supply
- *Not idolising, but... “Lost coins only make everyone else's coins worth slightly more. Think of it as a donation to everyone.”*

P2MS - Recent resurgence in use is for STAMPS protocol ([ref](#), [ref](#))

P2TR - A secondary issue

- Small but growing share of value
- Generally key-path spent → BIP-360?

# KEY OBSERVATIONS

~6.51M BTC (32.73% of total supply) is immediately QV

Address Reuse: 4.49M BTC → most of these funds can likely be moved? 

- T1000 show higher rates of reuse (30%) than all addresses (15-20%)
- T1000 account for ~2.73M BTC of the immediately QV

Fork exposure: 0.15M BTC → most of these funds can likely be moved? 

P2PK: 1.72M BTC, almost entirely coinbase TX → most likely won't move 

P2TR: 0.15M BTC, many non-economic UTXOs, key-path favoured

***Estimate: <2M BTC of the immediately QV  
funds can't/won't be ever moved***

**THANKS!**  
**&**  
**QUESTIONS?**